

Task 1: Understanding the users

Before you design and implement any piece of software, you need to have a detailed understanding of the people that will be using it. You need to know about their goals, their capabilities, and the conditions under which they will be using your software. Once you have this understanding, you can proceed with producing a design.

For this task, you will choose and identify your target user group, identify other stakeholders, and develop requirements about your chosen target user group as detailed below. Consider the steps as being iterative – i.e., as you iterate through these steps and gain more understanding about the domain and user's needs, you will be able to better describe your target user group and their requirements.

1. Identify a target user group:
 - Specific, knowable, and interesting to you
2. Choose 2-3 appropriate user research methods to understand your target user group, and also to identify and understand any other stakeholders:
 - Which user research methods did you choose to use? Why? (max 100 words)
3. Describe the results of your user research and provide visualisation of the analyses (max 500 words, figures do not count towards the limit).
 - What did you find from quantitative data analysis?
 - What did you find from qualitative data analysis?
 - Any unexpected / surprising results?

Support your findings with evidence. If you used interviews or questionnaires, include the questions as appendices. If you gathered data from articles, journals, papers etc. make sure you cite these.
4. Write a Persona illustrating your target user group, summarising what you know and learned about them (max 600 words).
 - Who are they, what did you learn about them, how do they relate to the interaction, etc.
5. Identify and describe the wider stakeholders (max 400 words)
 - Describe each stakeholder group identified, and their position in the context of the interaction. Who are they, what did you learn about them, how do they affect or are affected by the design/usage of the app, and so on.
 - If you find it helpful, feel free to create Personas for each/some of the stakeholder groups
6. Develop user requirements - identify and list all of the relevant user requirements that your design must meet (max 300 words)
 - What are the goals/tasks that the user must be able to perform? What are some of the attributes from the user's perspective that the app must have?
 - If you find it helpful, feel free to create use cases, scenarios or user stories to list the user requirements

NOTES:

Do not treat each section independently. Merge all parts into one coherent submission.

To get the full tick you need to provide ALL of the above (no item is optional).

SUBMISSION: Create a PDF document that contains the above user research data gathering and requirements development. Submit all of the above as a single PDF file titled Task1_GroupXY.pdf for the whole group via the moodle module page.

DEADLINE: 7 May noon - 12pm

- Uncategorized
 - Stakeholders
 - Windsurfers / Water sport
 - Lifeguards / RNLI
 - Weather provider
 - User research questions
 - What kind of weather do you look for before going windsurfing?
 - How do you currently check the weather?
 - Do you use multiple apps for checking the forecast? Why?
 - How in depth do you look at the forecast?
 - App features
 - Recommend sail sizes based on wind
- 1. We chose windsurfers as our target user group
- 2. User research methods
 - a. Observation (Ethnography)
 - i. Watching youtube videos, seeing what weather is preferable for windsurfers what they look out for before going out
 - ii. Easy to source data for observations allows us to see what data they find most crucial
 - b. Focus Groups (Same target user)
 - i. Modelled with claude, gives a
 - ~~c. Interviews (Semi-structured)~~
 - ~~i. Speaking directly with windsurfers to understand their behaviour around weather, what they would want to get out of weather app~~
- 3. User research results
 - a. Observation results

Windsurfing Weather - User Research

Target User Group

We decided to pick windsurfers as our target user group. Windsurfers are very sensitive to weather and rely on accurate weather forecasts to schedule and plan their excursions. Windsurfing is interesting to us as the need for specific, optimal weather makes them an ideal focus for a tailored weather app.

User Research Methods

The two user research methods we chose to use were observational data, collected from YouTube videos and content analysis of community forums, and focus groups where we solicited discussion from groups of windsurfers on how they use weather apps and what they want from them. We read forums online such as r/windsurfing on Reddit, and then noted down exact quotes by people, which we then classified into themes and the types of activity that people were doing.

User Research Results

Observations

Through observation we found 3 main results. Windsurfers online consistently reported that they relied on multiple weather apps for their data, they would cross reference apps like Windfinder, Windguru, and Windy. In one YouTube video (<https://www.youtube.com/watch?v=1Fvp37ZV2hc>) a windsurfer recommends to beginners, "I often hear the discussion which forecast is best. Is it Windfinder, is it Wind Guru, is it Windy? ... But that's a bit beside the point, you should compare all of them". Additionally, we found that many inexperienced windsurfers were often unsure what wind speeds were sufficient to sail in; on the Reddit community r/windsurfing we observed users frequently post screenshots of forecasts asking, "is this enough wind to go out?", indicating that raw data (knots, direction) is often hard to interpret without experience. Once more, we found from professional surfers that sail size is highly variable - many existing apps use simple formulae to recommend a sail size, however, in reality, the correct sail size depends on board type, fin size, the water current and sailor ability. The observation references and thematic analysis follow.

Website / Platform	User ID	Exact quote / evidence excerpt	Theme / Subtheme	Type of activity
Reddit / r/windsurfing	R1 — Vok250	"When you talk about wind speeds on this subreddit are you referring to the max gusts or the average?"	Confusion interpreting forecast data; average wind vs gusts	Beginner planning / understanding forecast
Reddit / r/windsurfing	R2 — Maxifloxacin	"Is this enough wind to learn or move?"	Beginners struggle to translate knots into action	Beginner decision-making
Reddit / r/windsurfing	R2 — Maxifloxacin	"I've been looking at the windfinder app... the app says 2-7 knots... and 3-8 knots sunday."	Users check forecasts around free time / day-off constraints	Trip planning
Reddit / r/windsurfing	R3 — abrandis	"anything above 5kts should be adequate to learn."	Beginner threshold; conditions depend on skill level	Learning conditions
Reddit / r/windsurfing	R3 — abrandis	"Also depends on other factors, sail size, board size and your weight."	Forecast data must be personalised by gear and body weight	Gear-aware interpretation
Reddit / r/windsurfing	R3 — abrandis	"use an app like Windy or windfinder to check forecasts and try to time your sessions"	Forecast apps used to time sessions	Forecast checking
Reddit / r/windsurfing	R4 — lysanderhawkley	"I windsurf on big boards in 8 knots, and smaller boards from 18 knots."	Suitable wind varies by board type	Equipment-dependent conditions
Reddit / r/windsurfing	R4 — lysanderhawkley	"Windsurfing starts to get really fun at around 18 knots."	Distinction between possible, learnable, and fun conditions	Session quality judgement
Reddit / r/windsurfing	R5 — kdjfsk	"the bigger problem is wind isn't constant, it tends to go +/- 4-5kn of the forecast"	Forecast uncertainty; forecasts need confidence/spread	Forecast reliability
Reddit / r/windsurfing	R5 — kdjfsk	"this could leave you stranded and needing rescue."	Forecast error can create safety risk	Safety / risk warning

Reddit / r/windsurfing	R6 — darylandme	"I am about 78kg and would use the 7.2 in about 17-23 knots. The 6.4 would be 20-26ish."	Sail size depends on wind range and rider weight	Sail selection
Reddit / r/windsurfing	R7 — bravicon	"12-16kt: 7.2 / 17-20kt: 5.6 / 21-25kt 4.7"	Users want wind-to-sail recommendations	Gear recommendation
Reddit / r/windsurfing	R5 — kdjfsk	"All the avid windsurfers i know including me have 6 or more sails."	Windsurfers own multiple sails; quiver management matters	Gear planning
Reddit / r/windsurfing	R8 — Severe_Tap9771	"not enough wind... mean[s] it's not enough wind for that equipment."	Raw wind speed is insufficient without equipment context	Equipment-aware forecast
Reddit / r/windsurfing	R8 — Severe_Tap9771	"we also have different boards and board types for different conditions."	Board type changes interpretation of same forecast	Gear / conditions matching
Reddit / r/windsurfing	R5 — kdjfsk	"I almost always pack two board[s]... Just in case the wind changes unexpectedly."	Users compensate for forecast uncertainty by overpacking	Trip preparation
Reddit / r/windsurfing	R9 — putin_vor	"Wind forecasts, no matter which website you use, are not generally reliable."	Distrust of single-source forecast	Forecast reliability
Reddit / r/windsurfing	R10 — kdjfsk	"Predict wind has the best UI... the tables have all the data at a glance"	Users value readable, glanceable data displays	App comparison / UI evaluation
Reddit / r/windsurfing	R10 — kdjfsk	"I've learned not to put too much faith in any of them."	Users distrust individual apps and models	Cross-checking behaviour
Reddit / r/windsurfing	R11 — unfknblvablen8	"Windguru.cz is the most comprehensive. Has everything you need basically."	Windguru perceived as feature-complete	App preference
Reddit / r/windsurfing	R12 — Tsambikos96	"only wind direction, and wind relative to the beach."	Direction must be shown relative to launch beach	Coastal context

Reddit / r/windsurfing	R12 — Tsambikos 96	"offshore wind... is extremely dangerous - especially for an inexperienced surfer."	Offshore warning is safety-critical	Risk warning
Reddit / r/windsurfing	R5 — kdjfsk	"beginners want onshore wind"	Beginner-safe conditions should be explained plainly	Beginner safety
Reddit / r/windsurfing	R13 — some_whe re_else	"Don't go out in offshore winds unless you have rescue boat cover."	App should flag offshore wind / rescue requirement	Safety / lifeguard relevance
Reddit / r/windsurfing	R14 — bravicon	"Flat water, closed bay. Wind 8-12kt, side or onshore, never offshore."	Beginner condition recommendation: flat water, moderate wind, safe direction	Beginner session planning
Reddit / r/windsurfing	R5 — kdjfsk	"even this is an estimate, because theres so many variables... your skills are a factor, too."	Sail calculators are limited; skill and gear matter	Sail-size calculation limits
Kitesurfing Dublin tutorial	Article author	"The 2 most important factors are: direction and strength."	Wind direction and strength are core weather variables	Forecast literacy
Kitesurfing Dublin tutorial	Article author	"Wind speed is the average speed of the wind... Wind gust is the max wind speed"	Users need average/gust distinction explained	Forecast interpretation
Kitesurfing Dublin tutorial	Article author	"The tides impact your sessions and it is important to know which tide you can expect"	Tide is part of session planning	Coastal planning
High Spirits Kitesurf tutorial	Article author	"It's always smart to compare Windguru with a second forecast."	Cross-checking multiple models/apps	Forecast comparison
YouTube — from your observation notes	Video narrator — add timestamp	"I often hear the discussion which forecast is best. Is it Windfinder, is it Wind Guru, is it Windy? ... you should compare all of them."	Users are advised not to rely on one forecast source	YouTube observation / forecast education

Focus Group

We organised a focus group with 4 regular windsurfers ranging from beginner/intermediate to advanced levels where we discussed their current weather app usage, the limitations of these apps, their weather requirements for windsurfing and how these might work inside an app. We used a semistructured approach, with guiding questions encouraging open discussion.

Focus group participant overview:

	User 1	User 2	User 3	User 4
Name	Sophie	Chris	Georgia	Benjy
Age	50-54	55-59	20-24	20-24
Estimated windsurfing level (1-10)	4	9	2	7
Estimated technical proficiency (1-10)	6	7	8	9

Thematic Analysis of the Focus Group:

Theme	Key Findings	Participant Comments
Condition Requirements	Ideal conditions are not universal; they are defined by the specific discipline and the athlete's current ability.	"What counts as a 'good' day is totally subjective. I'm probably looking for very different conditions, like less wind, than a pro would." - Georgia
		"I'm generally looking for about 15 knots and I don't really have a limit... I definitely don't bother with temperature if the wind is up." - Chris
Verification & Trust	High distrust of single-source forecasts leads to manual cross-referencing across multiple platforms.	"I never rely on just one source. I'm usually toggling between different apps and models to see where they overlap to get a general picture." - Chris
		"I check a couple of apps, just in they're saying totally different things. Maybe also check the webcam to see what others are doing." - Benjy
Hierarchy of Metrics	Wind (strength/direction) and tide are the "big three," while comfort (temp/rain) matters more to novices.	"Wind obviously determines if we can go out at all so is the priority, but if it's freezing or raining, I'm also much less likely to head out." - Sophie
		"The temperature only really plays a factor in choosing which wetsuit I take" - Benjy
Model Transparency	Advanced users prefer seeing the "spread" of different models rather than a single,	"I need to know where the models agree and disagree. If there's a large disparity between models then I'm more likely to be conservative with tide times and sail

	potentially wrong prediction.	sizes." - Chris
		"I want a consolidated view that makes it easy for me to flick between spots, but I still want the option to expand the technical data." - Benjy
Actionable Intelligence	There is a demand for "translated" data, such as recommended sail sizes and beach-relative wind directions.	"Often the recommended sail sizes are completely off, recommending a 2m sail when I would take a 4m, and this also changes depending on the board." - Chris
		"If it's a new spot, I don't know whether the direction is offshore, which would mean I can't go." - Sophie
		"As a beginner, having a recommended sail size also helps me gauge how windy it will really feel." - Georgia

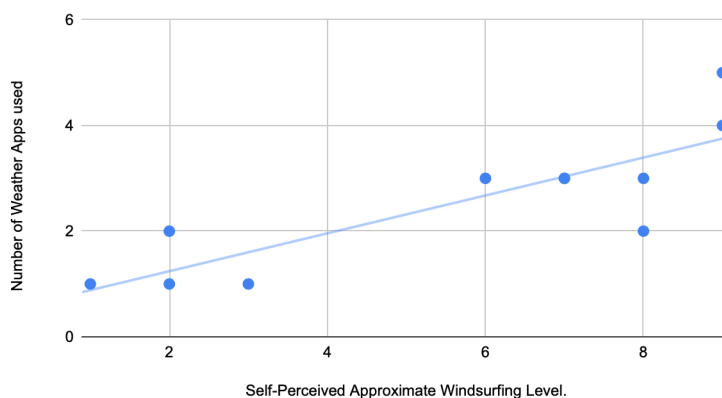
Key Observations from the Focus Group Thematic Analysis:

- There exists a correlation between a user's windsurfing ability and the complexity of the tools used. A successful product must cater to both by providing a general summary while allowing advanced sailors to view further metrics.
- All users shared similar technical proficiencies, which suggests the beginner's preference to more 'general' apps is primarily due to a lack of domain specific understanding. The app must be accessible by providing jargon explainers, or interpreting the data for the user.
- The manual process of app-hopping and data comparison is a significant barrier to beginner windsurfers and a time strain for advanced; consolidating data into a single app (both different types of data, and different models of the same data) will improve the current user journey.
- Beginners and intermediate users would benefit from risk assessment - this may involve sail size recommendations for a more understandable metric, or an outright risk warning for particular wind directions or states of tide (eg. offshore wind, or outgoing spring tide).

Quantitative data:

We gathered quantitative data using a questionnaire which was completed by members of the focus group and a number of other windsurfers. We received 11 responses, limiting the statistical strength of any conclusions but still providing useful insight into the requirements of different users. The resulting charts have been considered with regard to the user needs discussed in the focus group and observations to provide useful conclusions.

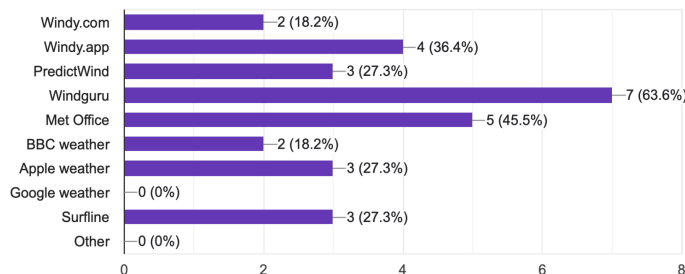
Windsurfing Level vs Number of Weather Apps Used



This graph shows the correlation suggesting more advanced windsurfers tend to use upwards of 2-3 apps to forecast the weather. This is a significant obstacle in the user journey, therefore, our app must combine both different data types and information from different models into a single overview.

What wind forecast apps do you currently use?

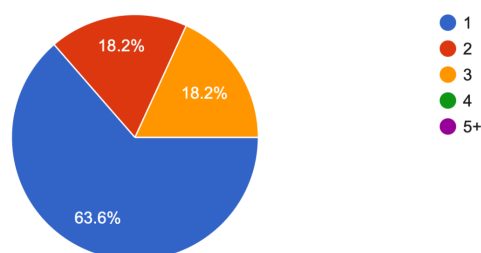
11 responses



This chart suggests that Windguru, a highly detailed, information-heavy site, is the most popular, while the more 'general' Met Office is still used by many. Surflife, used for accurate swell prediction, was used by 27% of respondents, indicating the need for swell forecast in the app.

How many boards do you own?

11 responses



This chart indicates that the majority of users have only 1 board, so our kit recommendations, while they still would benefit from taking into consideration board size, do not need to also recommend suitable boards for the conditions.

Examples of pre-existing windsurfing sail calculators



Windy.app provides a recommendation based on the wind prediction and pre-configured user weight. It doesn't account for ability, tidal conditions, or board size. Users found it frustrating that Windy.app's recommendations were always very specific and typically not a size they owned, making choosing a sail difficult when they found themselves between sizes. Additionally, advanced sailors found Windy.app's recommendations to regularly be too small for the conditions.

Windsurf Sail Chart

 KATANAGRANS Rider Weight Kg	Wind speed range, knots, Beaufort									Rider's skill level		
	kts	6-8	8-10	10-13	13-17	17-21	21-25	25-31	31-37	37+	Beginner	Intermediate
	BF	2	3	4	4-5	5	6	7	7-8	8+	Beginner	Intermediate
	Windsurf sail size, m²									Windsurf board volume, litres		
	50-60	7-9	6-8	5-6,5	4,5-5,5	3,3-4,3	2,8-3,7	2,5-3,5	2,3-3	> 2,7	145-160	95-108
60-70	8-10	7-9	6-7,5	5-6,2	4-5	3,3-4	3,1-3,7	2,5-3,5	> 3,0	160-175	108-121	
70-80	9-11	7-10	6,5-8	5-5,7	4,5-5,5	3,8-5	3,5-4,2	3-4	> 3,5	175-190	121-134	
80-90	10-12	9-11	7,5-9,5	6,5-8	5-6	4,5-5,5	4,2-4,7	3,5-4,3	> 4	190-205	134-147	
90+	12+	11+	9+	7,5+	5,5+	4,7+	4+	3,5+	> 4,5	205+	147+	

This table from karamarans.com contains board size and skill level considerations, however, the wind ranges are an entire force which can easily span two sail sizes, rendering it unhelpful any further than to confirm apre-established decision.



Target User Group Persona

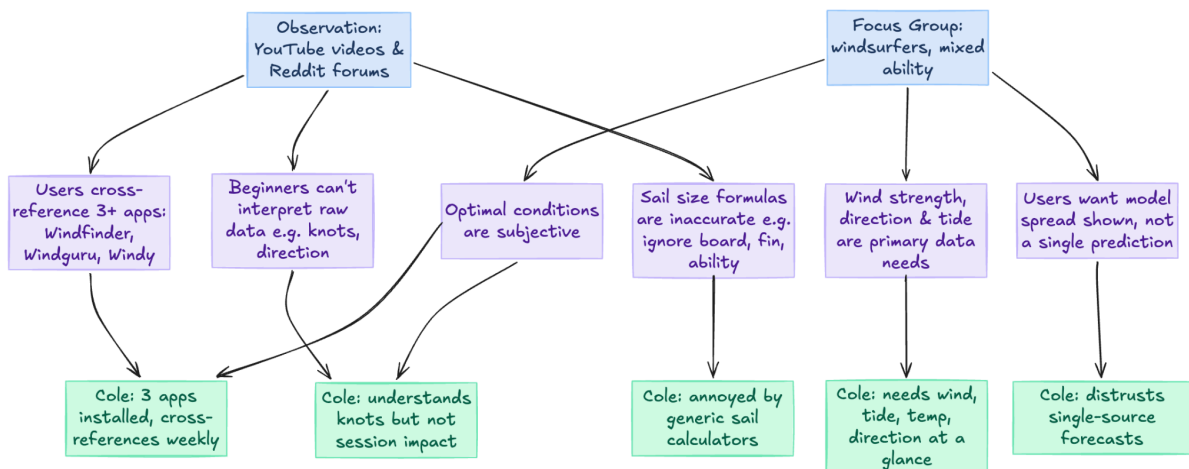
Name: Cole Dwarter

Occupation: Chef

Skill Level: Intermediate

Our target user group persona and an example of a primary stakeholder is Cole, an intermediate windsurfer who has been doing the sport for ~3 years. Because he lives 45 minutes away from the seaside, he relies on

accurate forecasts before deciding to drive down to the beach. Cole currently has three different weather apps installed on his phone (Windfinder, Windguru, Windy) which he has to toggle through every Thursday when planning his weekend. Cole owns multiple sails, but only has space for one of them in his car - it's important for him to know which sail to bring on the day.



Current problems:

- Cole is currently **skeptical of forecasts**: in the past weather apps frequently over or under predict the weather. i.e the app predicts 20 knots, but the beach is flat when he arrives. He does not trust single-source forecasts.
- He **struggles to interpret weather predictions** he does find; he can understand knots and wind direction but can't easily interpret what this means for his session today.
- He is annoyed by **basic sail-size calculators**, as these don't consider his board size, experience, rider-weight, or temperature, and often giving an arbitrary size rather than a sail size that he owns.

Cole is the core user of the app, he needs complex weather data from multiple sources to be turned into something he can quickly understand and use. When interacting with the app he doesn't want this data hidden from him, but presented in a manageable way alongside the simple insights that matter to him.

Cole will use the app's customisation features by inputting his specific gear (board type, fin size, and riding preferences) so that the app can make accurate sail size recommendations. Cole will open the app once or twice a week to plan his outings.

User Requirements

The core requirements of our app are based upon Cole being able to check on the weather from his home and make a binary decision on whether to go to the beach or not without switching between apps and forecasts. All requirements flow from this.

Forecast Aggregation and Uncertainty Display:

- The app must be able to pull from multiple sources and display them in a consolidated view. Cole needs to be able to see data like speed, direction, temperature along with the variance in prediction between the services.

Condition-Specific Display:

- The app should show all data points relevant to wind-surfing, e.g windspeed, direction, tide state, and temperatures, on a single page in a timeline format.

Gear Size recommendations:

- The app should allow Cole to input his quiver along with his skill level. Recommendations should include a primary recommendation along with when to size up or down to give Cole a better understanding.

Explanations and summary:

- The app must explain what the data-points mean and how much they factor into the overall conditions along with a traffic-light style indicator on whether the conditions are sailable along with a brief explanation.

Hazard Warnings.

- The app must prioritise showing red-flag conditions with explanations as to why these are so critical. This serves both safety (Cole) and regulatory concerns (RNLI / lifeguard stakeholders).

Performance Requirements:

- Forecasts must load within a reasonable time on a mobile connection and the interface must be usable one-handed on a phone screen since this will be primarily how Cole uses the app.

Stakeholders

Windsurfers (primary stakeholders).

These people are the target users of the app. More detailed information about them can be found in the Target User Group, User Research Results and Target User Group Persona sections of this document.

Data and infrastructure providers (secondary stakeholders).

These entities provide the raw GRID files and satellite data, and include the companies selling anemometers and smart buoys that supply essential ground truth data. Cloud infrastructure may be required for handling high-resolution weather models as these require significant computational power.

Commercial and monetisation partners (tertiary stakeholders).

Equipment manufacturers and retailers may be integrated with targeted advertising, potentially weather based. Insurance companies have a vested interest in apps that promote safety and provide conservative decision-making to reduce maritime claims. Kitesurfing schools or sailing clubs may want to be a pinned spot to drive foot traffic.

Existing weather and wind forecast apps (tertiary stakeholders).

Our main competitors may be affected by us as some of their current users that are also our target audience may switch to our app. They also affect us in the way that we need to ensure that our app significantly differs from them and has a unique selling point.

The apps that we take into consideration are:

- Weather prediction apps:
 - Met Office
 - BBC Weather
 - AccuWeather
- Wind prediction apps:
 - Windy.com
 - Windy.app
 - PredictWind

These apps are listed on Apple's App Store as the most downloaded apps in their respective categories as of May 2026.

Designers and developers (facilitating stakeholders).

This is us. We are responsible for translating complex, high-resolution multi-model data into an intuitive interface that windsurfers can easily understand. Our core focus includes ensuring system reliability and building the necessary architecture to seamlessly integrate the data providers and commercial partners mentioned above.

Task 2: Designing a Lo-fi Prototype

For this task, using the understanding you now have about your users and their requirements, produce ONE weather app design that serves the specific needs of the target user group you have identified in Task 1. You must ensure that your design(s) are informed by the results of the user research and requirements development you conducted previously.

Consider the steps below as being iterative, in particular 3 and 4. As you create the visual design, consider how the user will interact with the app. As you create the interaction design and storyboards, consider how the visual design supports that and make any needed changes to the visual design.

1. Start with brainstorming about the possible designs and solutions:
 - Consider which activities the user will want to do
 - Consider how your designs will support your target user group in their activities.
 - Consider how you will create novel and usable interactive designs.
 - If helpful, identify a central design concept for your app, or develop app-specific principles to guide your design
2. Decide whether the design is for a phone or tablet. Explain why your target user group would use one over the other (max 100 words).
3. Detail the **visual design** by focussing on the layout, functionality, and aesthetics of your lo-fi prototype. Illustrate clearly and in detail, your ideas and how your design decisions link to what you know about your target user and the other stakeholders. Create screen layouts (i.e., a visual sketch or mock-up of the appearance) illustrating the available screens of your app (3-5 screen layouts).
 - You can sketch this manually (using pencil and paper, cutting and pasting, and other similar techniques), or with digital means (using e.g. Figma, Balsamiq, Inkscape, Illustrator etc).
4. Detail the **interaction design** by focusing on which tasks and activities the user will want to do in the app, and how they will interact with it. Create storyboards illustrating several such key flows through the app and the user's interaction with it. (3-5 storyboards)
 - For example, consider the user wanting to "find today's temperature". Design the storyboard thinking about: What does the user have to do for each step? What do they have to click/tap/swipe/etc.? What happens next?
 - You can sketch this manually (using pencil and paper, cutting and pasting, and other similar techniques), or with digital means (using e.g. Figma, Balsamiq, Inkscape, Illustrator etc).
 - Aim to reuse and work with the visual screen layouts you designed above
5. Plan how you will go from lo-fi design to hi-fi design and implementation, detailing the following: need for external libraries and/or additional tools, distribution of tasks between team members, and potential risks and mitigation strategies (150 words max).

NOTES:

Do not treat each section independently. Merge all parts into one coherent submission.
To get the full tick you need to provide ALL of the above (no item is optional).

SUBMISSION: Create a PDF document that contains the above mentioned details of your lo-fi prototype: any design concepts / principles developed, detailed screen layouts/sketches, storyboards and plan for hi-fi implementation. Submit all of the above as a single PDF file titled Task2_GroupXY.pdf for the whole group via the moodle module page.

DEADLINE: 14 May noon - 12pm

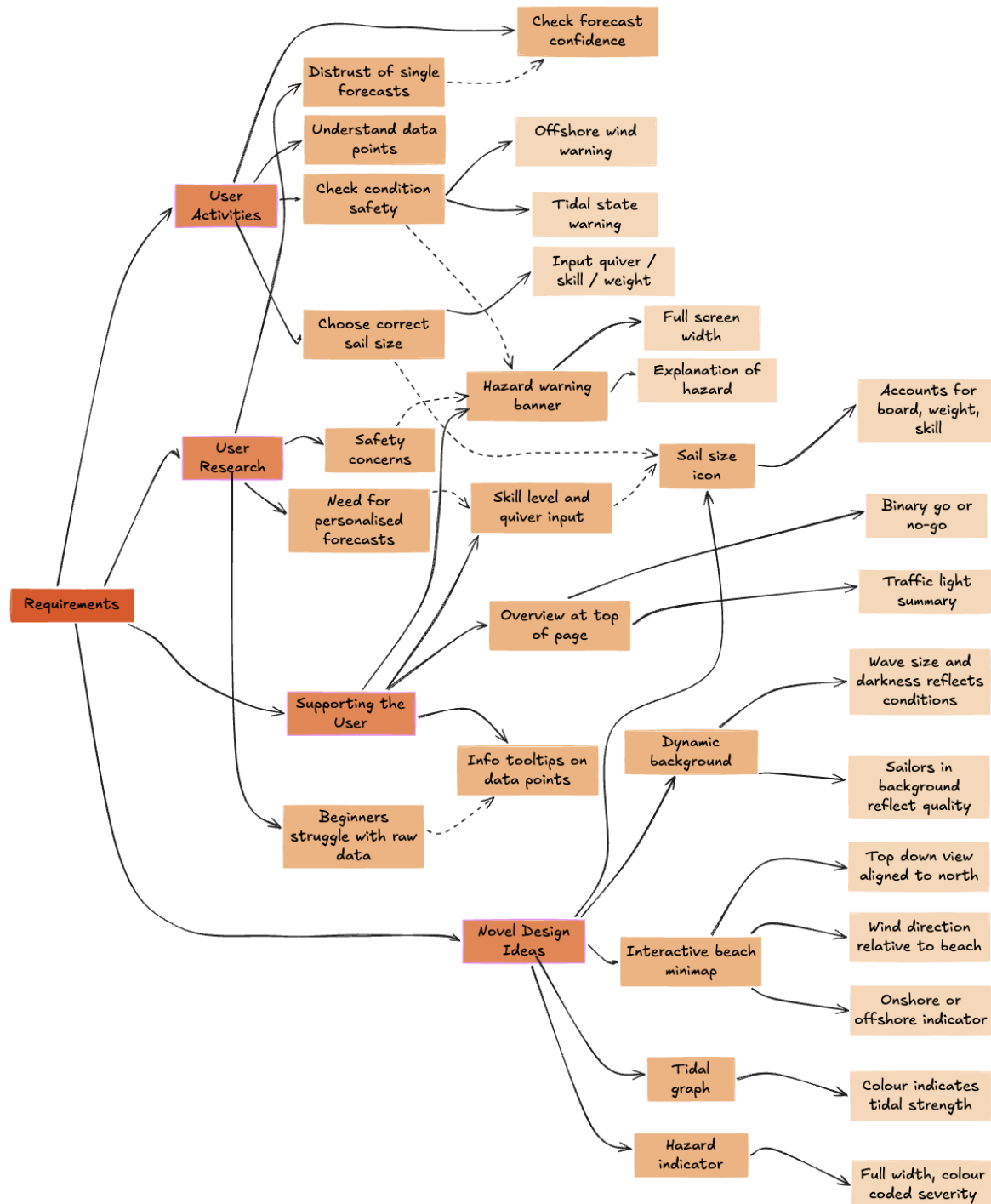
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1. Brainstorming

- What activities will the user want to do
 - Windsurfing
 - Check if the conditions are safe
 - Check what sail he should bring
 - Check how confident the prediction is
 - Understand what certain data means
- How will designs support target user group
 - Central component at top of page with overview
 - Hazard warning banner to show whether the conditions are safe
 - Ability to input skill level, quiver, etc.
 - Information tabs to show what datapoints actually mean
- How will we create novel and usable designs
 - Background of app changes with conditions (e.g larger, darker waves when conditions are rocky)
 - Arrow showing wind direction (maybe relative to beach)
 - Background showing current conditions.
 - Use static beach icon and rotate relative to direction of beach and keep direction relative to north - or use satellite imagery for beach.
 - Colour of tidal graph indicates strength of tide.
 - Number of sailors in the background shows how good the conditions are
 - Changing icon corresponding to sail size recommendation
 - Hazard level should be the whole width of the screen
 - Interactive circle illustration for beach and wind direction

- Circle center piece that look like a 'top view' / 'COD minimap' of the beach, you can see the beach and water aligned with the north direction.
- Illustration has annotations of wind and wind direction, as well as water conditions

Windsurfing Weather - LoFi Prototype



Brainstorming

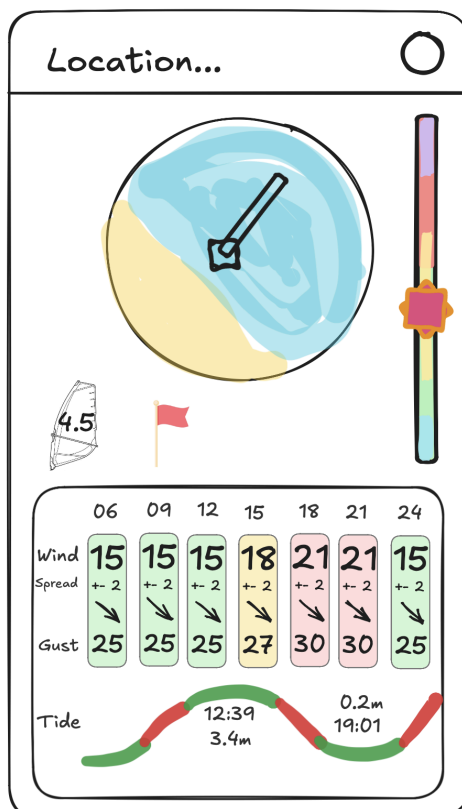
We brainstormed by linking our initial research to different possible design features for the lo-fi prototype. The main requirements from our research were to check whether conditions are safe, understand confusing forecast data, suggest a

recommended sail size, and judge the forecast's reliability. We came up with ideas such as a traffic light go/no-go summary, prominent warnings for hazards, info tooltips, and an interactive beach minimap showing wind direction, with the objective of increasing accessibility for beginners and intermediates whilst still providing a full suite of data for advanced users.

Visual Design

The design of our system is optimised entirely for mobile usage rather than tablet usage as although our primary users and Cole are likely to first check the conditions at home, where they may have access to a tablet, it is also likely that they check again both on the way to the beach and at the beach. Due to the limited scope of the data that needs to be visible at the first glance, it is not essential to have a larger screen as we are able to compress the necessary information into suitably sized widgets on a mobile device. Therefore, we have decided to make our implementation exclusively for mobile devices.

Possible solutions:



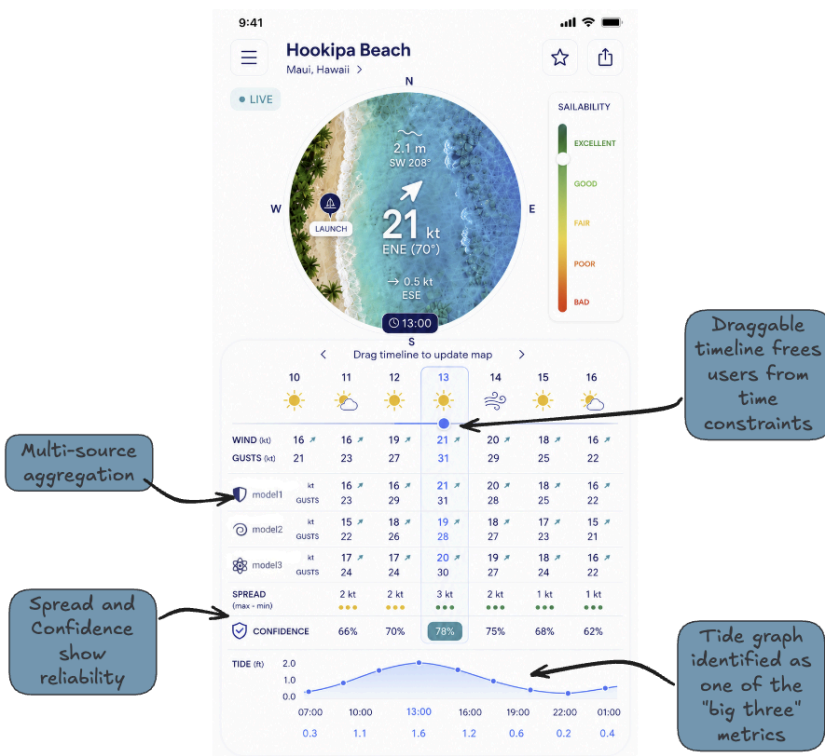
Concept prototype: we drew this to formalise ideas into a basic design, focusing only on key features such as the central map with data, conditions rating, and scrollable hour-by-hour forecast. We also considered the option of widgets, although these have been simplified to just icons in the concept design. Upon receiving feedback from the group and considering how the design satisfies the user requirements, we made incremental changes before reaching our final low-fi design (view next section). The most notable incremental changes were: scrollable widgets containing varying types of information, expanded views to encompass all model data into a single page.

Low-fi Design Screen Layouts

Main page:



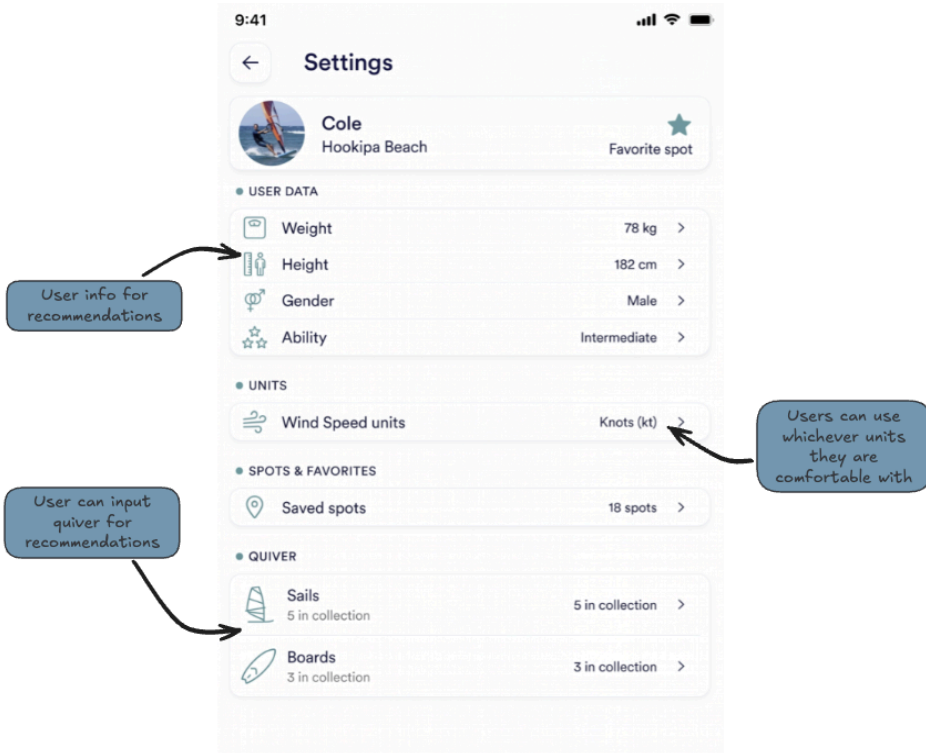
Expanded model view:



Locations Page:



Settings Page:



User Requirement Analysis

Main page:

UI Element	Satisfied User Requirement
"Sailability meter"	Explanations and summary: Provides a traffic-light style indicator on whether the conditions are suitable.
Beach map with wind speed and direction	Condition-Specific Display: Shows windspeed and direction on a single page.
"Safety" widget	Hazard Warnings: Addresses the need to display red-flag conditions with explanations.
Daily timeline showing wind and tide data	Condition-Specific Display: Displays relevant data points in a timeline format.
Bottom panel with temperature	Forecast Aggregation and Uncertainty Display: Satisfied the need to see temperature data in a consolidated view.

Extended view page:

UI Element	Satisfied User Requirement
Draggable timeline	Condition-Specific Display: Shows conditions for various times thus freeing users from time constraints
Tide graph	Explanation and Summary: Gives a simplified view of how the tide will move during the day
Multi-source display	Forecast Aggregation and Uncertainty Display: Shows multiple data sources to give Cole a better understanding of predictions
Confidence Value	Forecast Aggregation and Uncertainty Display: Percentage value gives Cole a sense of how well all sources agree with each-other with low values leading to uncertain conditions
Spread Value	Forecast Aggregation and Uncertainty Display: Shows Cole the

	variance in the different model's predictions.
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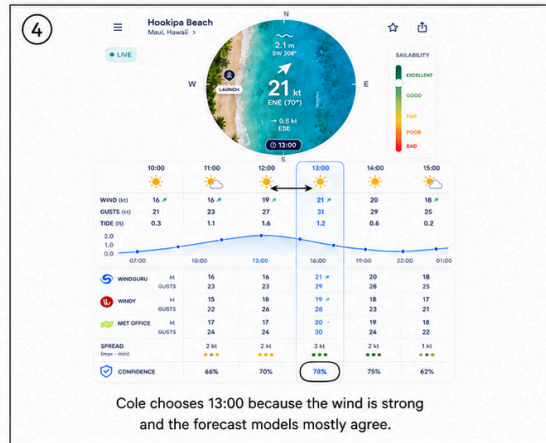
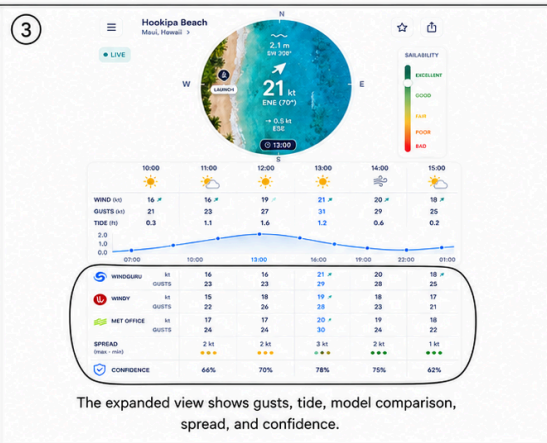
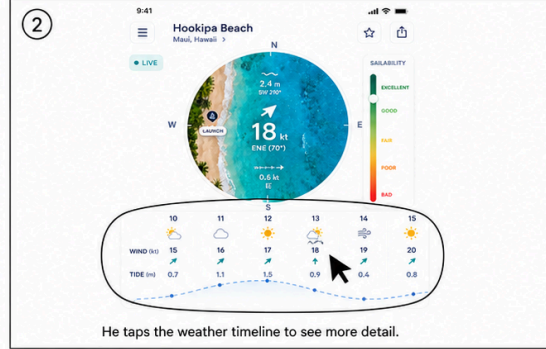
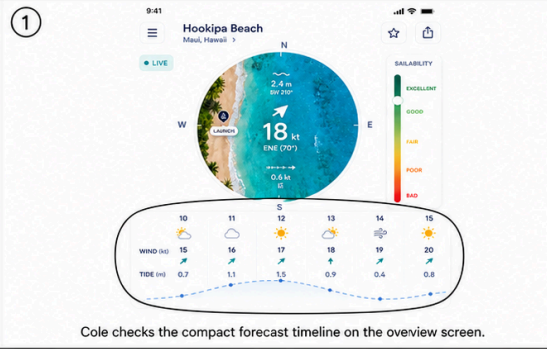
Locations page:

UI Element	Satisfied User Requirement
Search bar	Condition-Specific Display: Allows users to find conditions of a specific beach.
Edit favourites button	Performance requirements: Allows a user to quickly select which beach they want quick access to.
Distance to beach metric	Condition-Specific Display: Allows users to factor in travel times to their research, and also confirm they're checking the correct location.

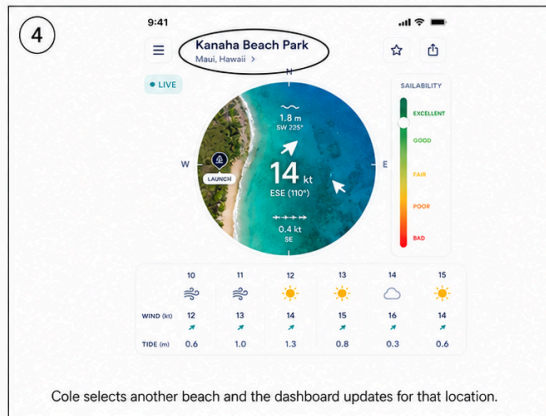
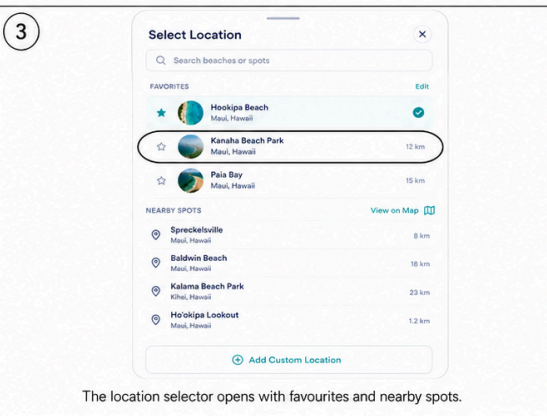
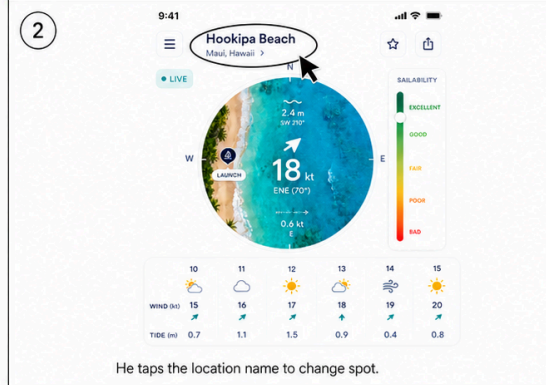
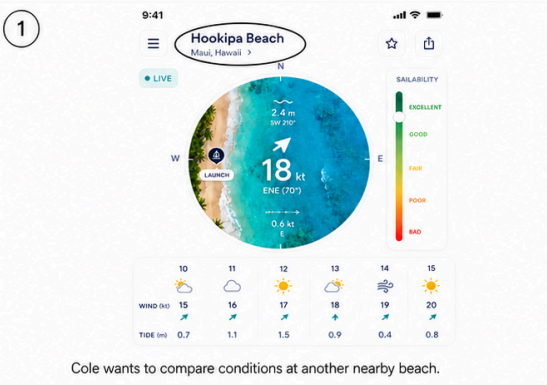
Settings page:

UI Element	Satisfied User Requirement
"User data" section	Gear Size recommendations: Allows the user to input their skill level and attributes to calculate personalised recommendations.
"Quiver" section	Gear Size recommendations: Allows the user to input their specific quiver (gear) so the app knows what equipment they own.

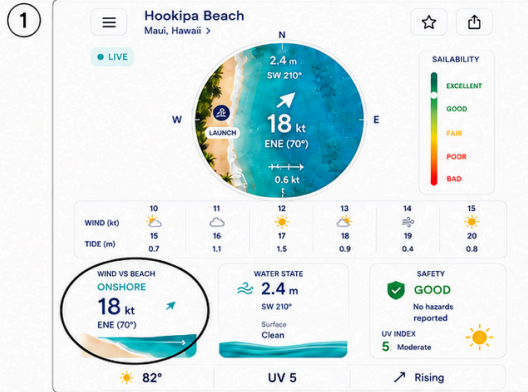
Storyboard 1: Expand Forecast Timeline



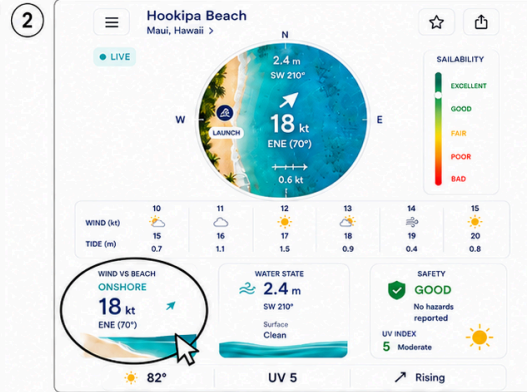
Storyboard 2: Change Location



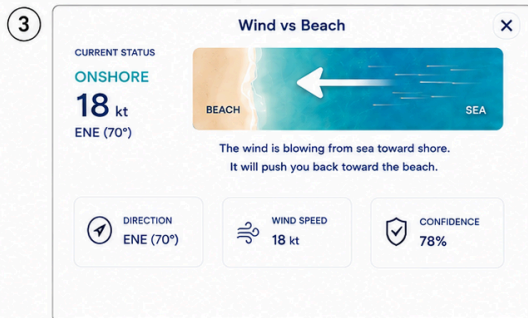
Storyboard 3: Check Wind Direction vs Beach



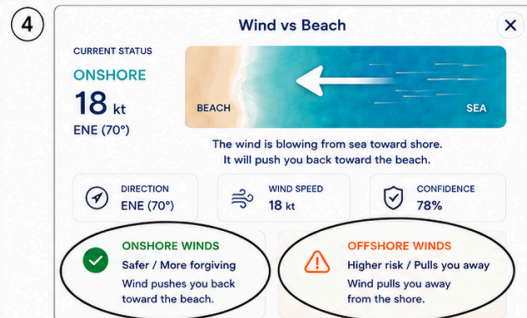
Cole sees that the app classifies the wind as onshore.



He taps the widget to understand the wind direction.

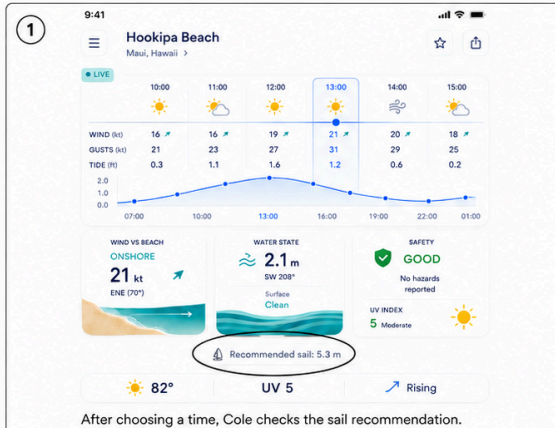


The expanded view shows the wind direction relative to the beach, not just compass degrees.

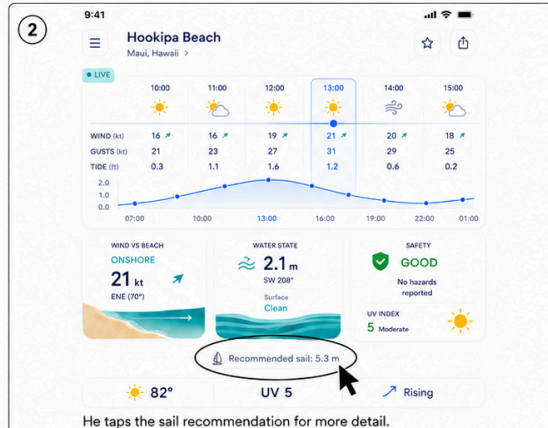


Cole understands why the direction is safer for his ability level.

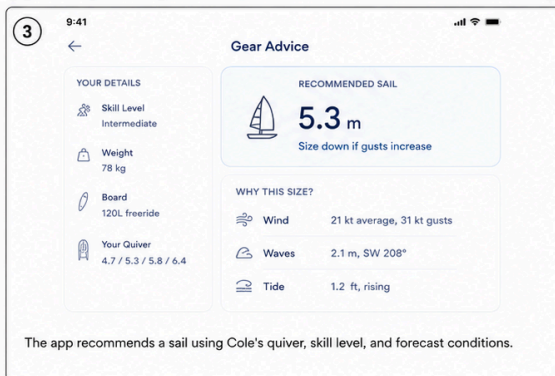
Storyboard 4: Get Sail Recommendation



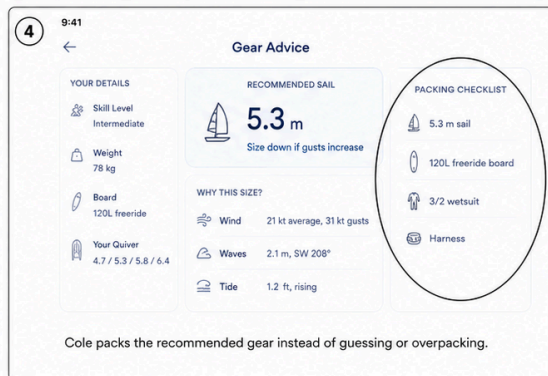
After choosing a time, Cole checks the sail recommendation.



He taps the sail recommendation for more detail.



The app recommends a sail using Cole's quiver, skill level, and forecast conditions.



Cole packs the recommended gear instead of guessing or overpacking.

Low-fi to High-fi implementation details

Data Collection

DATA REQUIREMENT	SOURCE	MODELS / PARAMS
Wind data	Open-Meteo	GFSECMWF IFSICONCopernicus ERA5 wind_speed_10m, wind_direction_10m, wind_gusts_10m
Swell data	Open-Meteo	ECMWF WAMNCEP GFS-WaveERA5-Ocean wave_height, swell_wave_height, swell_wave_period, swell_wave_direction
Tide data	WorldTides	FES2014TPXO9 heights, extremes, datum
Nearest tidal station	WorldTides	lat/lon lookup Returns station name, distance, and ID
Beach direction	Manual	coastline geometry Derive angle from coastline coordiantes near beach point
General weather	Open-Meteo	GFSECMWF IFSICON temperature_2m, precipitation, cloud_cover, uv_index, visibility
Optimal conditions	Computed	Calculated from all above sources

Library and Framework Decisions:

Front-end	Flutter	Provides cross-platform integration with a single codebase and highly customisable for implementing unique UI features.
Back-end	AWS-lambda	Serverless approach focuses on event-driven tasks, limiting API calls without overhead of a full back-end.
Network	Dio	HTTP client for Flutter which supports global configuration and allows us to manage concurrent API calls.
Local storage	Hive	No-SQL database to store quiver and user details on device and provide fast access.

Plan for Team Design Approach:

Access control: We will store our codebase on GitHub and thus use Git to handle merges.

Division of responsibility: We intend to divide responsibility for development corresponding to the areas we each focused on in the first two tasks.

Development approach: Since the majority of time will be spent developing the front end, we will assign different parts of the front end to each of us once we have established the back-end interface and allow one person to focus on the data processing.

Risks and Solutions:

Risk	Solution
Forecast is incorrect.	Provide transparency in data by offering a full model expanded view; include disclaimer of forecast.
Beach direction data is inaccurate.	Remind the user to check the direction when they arrive at the beach; include 'estimate' disclaimer in relative wind direction widget.
One or more models are vastly in disagreement causing conflicting forecasts.	Expanded view allows users to identify this; use confidence rating to review the consistency and agreement of models.

Risks include inaccurate weather data, which is, of course, very dangerous to windsurfers - to mitigate this, we need to provide information from multiple APIs so that one incorrect prediction is less likely to cause someone to windsurf in a dangerous way. Also, we need to provide a clear disclaimer that we take no responsibility for any poor decisions made as a result of predictions by the app, whether correct or not.

Task 3: Implementing a Hi-fi Prototype

Develop the hi-fi interactive prototype of the weather app you designed in Task 2. For simplicity of implementation, you will code, run and test your weather app on desktop/laptop. Use mouse input to emulate touch input. Build the interface using Java and/or its variants (e.g., Java FX, Android Java, Flutter, or similar) as and where appropriate (whether you choose to use a GUI Designer/Scene Builder or not is up to you. The focus here is on the functionality, usability and aesthetics of the user interface – do not spend time developing back end programs such as database systems. According to the needs of your target user group, choose an appropriate free weather provider API to develop your weather app.

In Task 3, we are interested in seeing both the demonstration of your programming skills and how well your implementation fits the design and the requirements you identified in Task 1. Important aspects you should address:

- Interactivity - how well the system interacts (fast and smooth transition between screens);
- Clarity - how clear your hi-fi design is
- Aesthetics - appropriate and pleasing use visual design features (e.g. background/foreground, colours, fonts, icons, and grouping of information)
- Structuring your code - good Model-View structure
- Supporting multiple screen ratios - using more sophisticated layout than absolute positioning
- Advanced interaction and programming - e.g. developing your own widgets vs. inheriting from AWT/Swing; building UI from scratch without a GUI Designer/Scene Builder
- Advanced graphic design - e.g. professional quality graphics elements, layout, and animations

Consider the following:

- Your hi-fi prototype should be interactive, intuitive and self-explanatory – i.e., a user should not need a manual to be able to use it
- Your code should be legible to externals – i.e., you should comment your code thoroughly
- Your code must acknowledge any elements re-used from your previous work(s) or any external code used
- Your code should run without you being present – i.e., you should provide a Readme file that explains how to run your weather app, what libraries you have used, and if it runs on any Windows / Mac / Linux computer etc.

Besides implementing the hi-fi prototype, also prepare a brief document covering the following:

1. Conduct one form of usability evaluation on your prototype (e.g. usability testing, heuristic evaluation, cognitive walkthrough). Describe which method you used, what you learnt from the evaluation, and any details you changed in the design as a result of the evaluation (500 words max)
2. Did you need to deviate from your lo-fi design in any other way? If so, why? (150 words max)

SUBMISSION: Submit your code and supporting files electronically as a SINGLE zipped folder title Task3_GroupXY.zip for the whole group via the Moodle module page, which contains the following:

- All source code files, and any IDE files needed to compile and run your code
- A executable file (e.g. jar file), if that makes running your prototype easier

- Readme_GroupXY.txt which explains how to run your code and what libraries are used
- Changes_GroupXY.pdf which explains your usability evaluation and any deviations from your design with rationale for the changes

Deadline: 26 May noon - 12pm

Windsurfing Weather - HiFi Prototype

Conduct one form of usability evaluation on your prototype (e.g. usability testing, heuristic evaluation, cognitive walkthrough). Describe which method you used, what you learnt from the evaluation, and any details you changed in the design as a result of the evaluation (500 words max)

Did you need to deviate from your lo-fi design in any other way? If so, why? (150 words max)